

Cambridge (CIE) IGCSE Maths
Paper 3: Calculator (Set D)
Core

Friday 25 April 2025

Afternoon (Time: 1 hour 30 minutes)

Total marks

/ 72

Instructions

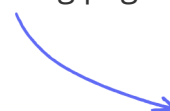
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You will need geometrical instruments.
- You should use a scientific calculator where appropriate.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- For π , use either your calculator value or 3.142.

Materials

- [List of formulas](#)

Scan here to mark your mock exam

or visit the mock exams landing page for this course



- 1 The number of cars parked in a car park at 9 am is recorded for 10 days.

124 130 129 116 132 120 127 107 118 114

Complete the stem-and-leaf diagram.



Key: 12|3 represents 123 cars

(2 marks)

2

8	15	18	33	39	41	51	57	60	81
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From this list, write down

i) a factor of 54,

[1]

ii) a multiple of 19,

[1]

iii) a prime number.

[1]

(3 marks)

3 Write down the reciprocal of 2.

(1 mark)

4 Work out $48 \div 3 - 5 \times 2$

(1 mark)

5 Find the value of

i) $\sqrt[3]{512}$

[1]

ii) $\frac{6^8}{2^6}$

[1]

iii) 7^0

[1]

(3 marks)

- 6 Work out the difference in temperature between -6°C and 5°C .

..... $^{\circ}\text{C}$

(1 mark)

- 7 In Tokyo, Jeremy buys a bracelet for 2050 yen.
The exchange rate is 1 yen = \$0.0125 .

Calculate the price of the bracelet in dollars.
Give your answer correct to the nearest dollar.

\$

(2 marks)

8 $A = \frac{1}{4}bc^2$

Calculate the value of A when $b = 3$ and $c = 6$.

(2 marks)

- 9 **Using a ruler and pair of compasses only**, construct a triangle with sides 5cm, 8cm and 10 cm.
Leave in your construction arcs.

(2 marks)

10 Calculate.

$$\frac{16.379 - 0.879}{4.2} \times 1.241$$

Give your answer correct to 2 significant figures.

(2 marks)

11 (a) On any day, the probability that Marcus will get a seat on the school bus is 0.93 .

Write down the probability that he will **not** get a seat on the school bus today.

(1 mark)

(b) There are 200 school days in a year.

Work out the expected number of days in a year that Marcus will **not** get a seat.

(1 mark)

12 (a) A circle has a circumference of 56mm.

Work out the radius of this circle.

.....mm

(2 marks)

(b) Calculate the area of a circle with radius 12cm.

..... cm²

(2 marks)

13 At Donville station the number of tickets sold each day is recorded for seven days.

104	18	72	31	27	45	60
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Find

i) the range,

[1]

ii) the median,

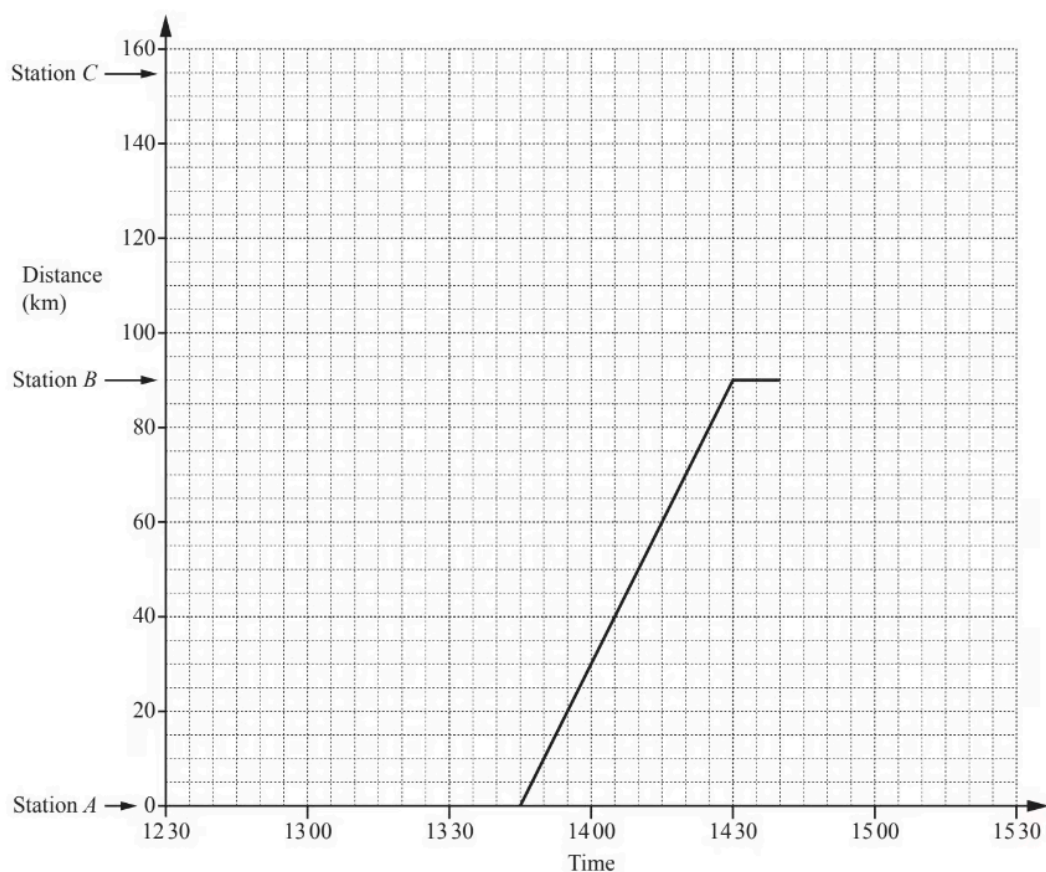
[2]

iii) the mean.

[2]

(5 marks)

14 (a) The travel graph shows part of a train journey between station *A* and station *C*.



i) Calculate, in km/h, the speed of the train between station *A* and station *B*.

.....km/h [2]

ii) The train leaves station *B* at 14 40.

For how many minutes did the train stop at station *B*?

.....min [1]

iii) The train travels at a constant speed between station *B* and station *C*, arriving at 15 20.

Complete the travel graph for the journey between station *B* and station *C*.

[1]

iv) On which part of the journey was the train travelling faster?

Between station and station [1]

(5 marks)

(b) Another train leaves station *C* at 12 45. It travels to station *A* at a constant speed of 62km/h without stopping at station *B*.

i) Work out how long, in hours and minutes, this journey takes.

..... hmin [2]

ii) Write down the time this train arrives at station *A*.

[1]

iii) On the grid, show the journey of this train.

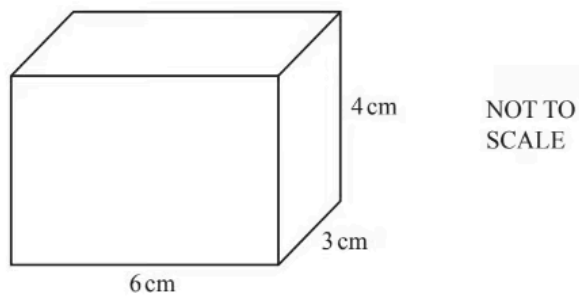
[1]

iv) Find the distance from station *A* when the two trains pass each other.

..... km [1]

(5 marks)

15 The diagram shows a cuboid.



i) Work out the surface area of this cuboid.

..... cm^2 [3]

ii) Work out the volume of this cuboid.

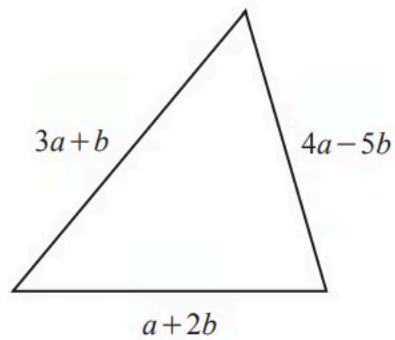
..... cm^3 [2]

iii) Write down the dimensions of a different cuboid that has the same volume as this cuboid.

..... cm by cm by cm [1]

(6 marks)

16



NOT TO
SCALE

Find an expression for the perimeter of this triangle.
Give your answer in its simplest form.

(3 marks)

17 A sphere has radius x cm. The volume of the sphere is 1000 cm^3 .

Calculate the value of x .

[The volume, V , of a sphere with radius r is $V = \frac{4}{3} \pi r^3$.]

$x = \dots\dots\dots$

(3 marks)

- 18 (a)** Gabriela designs the seating layout for a new theatre.
There are three sections of seats, A, B and C.

Section A has 152 seats.

Section B has 171 seats.

Section C has 57 seats.

Write down and simplify the ratio of the number of seats in each section A : B : C.

A : B : C = : :

(2 marks)

- (b)** For a concert in the theatre, the ticket prices are in the ratio

A : B : C = 9 : 7 : 4.

A ticket for Section C costs \$6.

- i) Show that a ticket for Section B costs \$10.50 .

[1]

- ii) Find the cost of a ticket for Section A.

\$ [1]

- iii) The table shows the number of tickets sold in each section.

Section	Number of tickets sold
A	120
B	136
C	30

Calculate the total amount received from the ticket sales.

\$ [3]

iv) The concert costs \$4500 to organise.

Calculate the amount received from the ticket sales as a percentage of the \$4500.

[1]

(6 marks)

19 Gino invests \$6000 for 5 years at a rate of 1.2% per year compound interest.

Calculate the value of his investment at the end of the 5 years. Give your answer correct to the nearest dollar.

\$

(3 marks)

20 At a bus stop

- a red bus arrives every 18 minutes

and

- a blue bus arrives every 24 minutes.

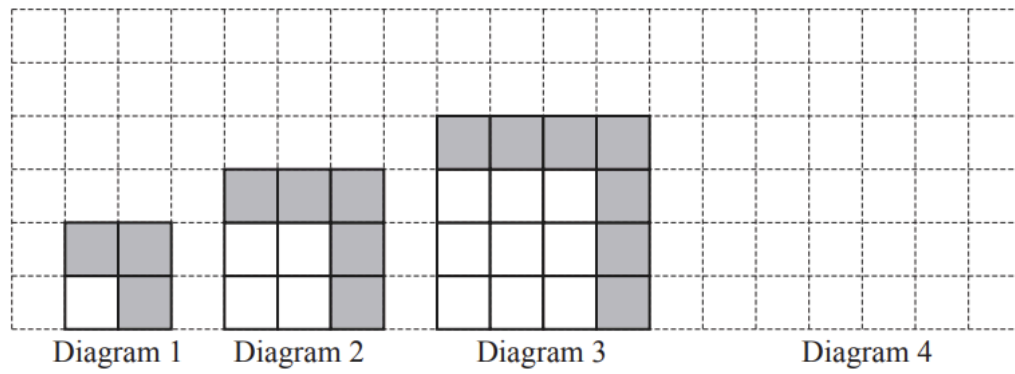
At 1047 a red bus and a blue bus arrive.

Find the next time when a red bus and a blue bus arrive together.

(3 marks)

21 The grid shows the first three diagrams in a sequence.

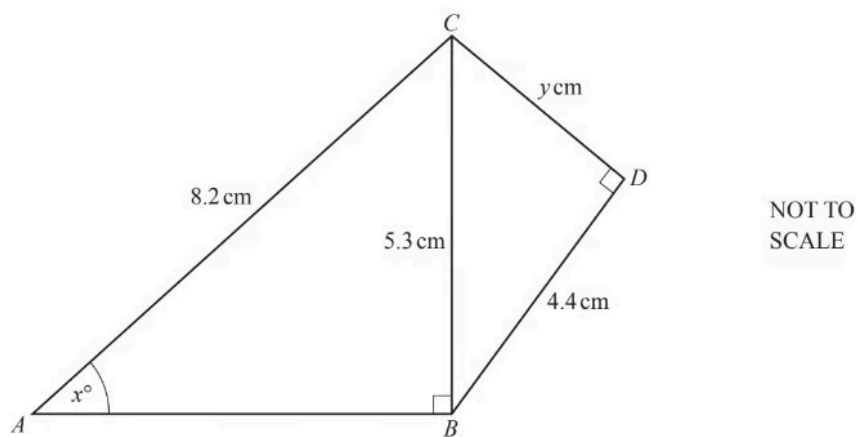
Each diagram is made using small squares that are white or grey.



On the grid, draw Diagram 4.

(1 mark)

22 (a)



Triangles ABC and BCD are both right-angled triangles.

Calculate the value of y .

$y = \dots\dots\dots$

(3 marks)

(b) Calculate the value of x .

$x = \dots\dots\dots$

(2 marks)